

Connections of Evolvability to Correlational SQ Learning

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References

- ▶ *Evolvability*, Valiant (2009)
- ▶ *A Complete Characterization of Statistical Query Learning with Applications to Evolvability*, Feldman (2009)
- ▶ Talk by Feldman on *Distribution-Independent Evolvability of Linear Threshold Functions*, COLT 2011
- ▶ Talk on *Quantitative Model of Innovation in Evolution* by Valiat (2014)

Similarities between learning and evolvability

Learning

- Goal is to predict an hypothesis of some process based on large set of possible unseen examples
- Learn from past experiences and observations; not optimal to model the entire globe
- Use examples of hypothesis

Evolvability

- Goal for organisms is to have good responses in many different conditions, seen or unseen
- Same as learning
- Feedback from the environment (Darwin theory - try numerous mutations/variations to current realization; best one survives)

Valiant's model of evolvability is that of restricted form of learning.

Example

Expression of a gene in the DNA regulated by *transcription factors* (TF). They look at the environment factors; other proteins and then determine how much protein will be produced by the gene.

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Then, the question would be - how do regulations mechanisms evolve?

Evolution Algorithm

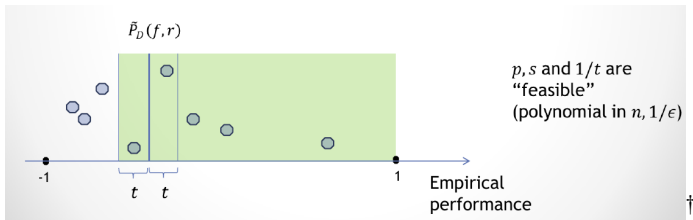
- ▶ R - representation class of functions over domain X .
Example: all linear thresholds over $\mathbb{R}^n$.
- ▶ M - randomized mutation algorithm that given $r \in R$ outputs $r' \in R$
 - ▶ Efficient: poly in $\frac{1}{\epsilon}, n$

Selection

- ▶ Fitness: $\text{Perf}_D(f, r) \in [-1, 1]$, where f is some ideal function;
 r function computed by our representation
 - ▶ Correlation $E_D[f(x)r(x)]^*$
- ▶ Run $M(r)$ p times to get $R = \{r_1, r_2, \dots, r_p\}$
- ▶ Estimate $\forall r_i \in R, \widetilde{\text{Perf}}_D(f, r_i) = \frac{1}{S} \sum_{j \leq S} r_i(x_j) f(x_j)$

p and S should be poly in $n, \frac{1}{\epsilon}$

*CSQ is a special case of SQ model



†Image src: Evolvability of Linear Threshold Functions, Vitaly Feldman

Evolvability (formally)

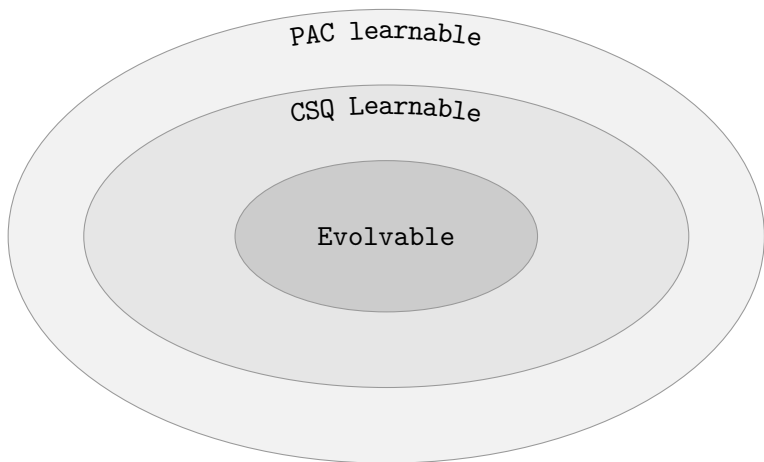
Class of functions C is evolvable over D if exists an evolution algorithm (R, M) and a polynomial g s.t.

for every $f \in C, r \in R, \epsilon > 0$ and a sequence $r_0 = r, r_1, \dots$ where $r_{i+1} \leftarrow \text{select}(R, M, r_i)$, it holds $\text{Perf}_D(f, r_{g(n, \frac{1}{\epsilon})}) \geq 1 - \epsilon$ w.h.p.

Evolvability Oracle

- ▶ Send oracle h_1
- ▶ Oracle returns $v_1 = \widetilde{\text{Perf}}_D(f, h_1)$ by using S fresh examples
- ▶ Send oracle h_2
- ▶ Oracle returns $v_2 = \widetilde{\text{Perf}}_D(f, h_2)$ by using S fresh examples
- ▶ ...
- ▶ ...

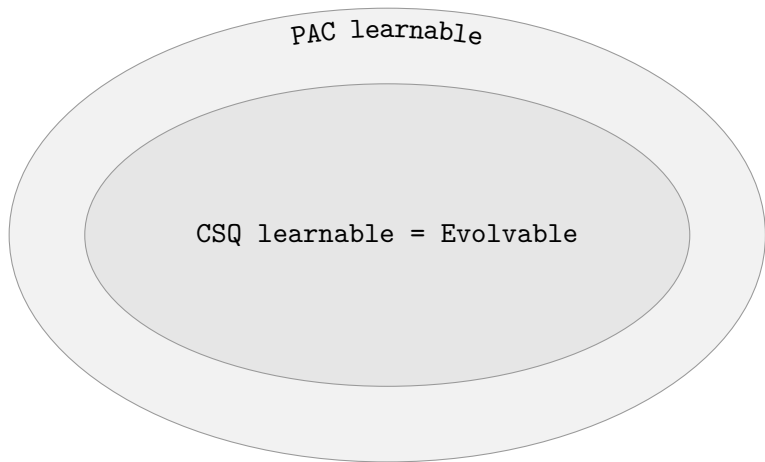
Big Picture



Equivalence with CSQ

Correlational Statistical Query:

- Send oracle query h
- CSQ oracle responds with $|v - \widetilde{\text{Perf}}_D(f, h)| \leq \tau$, and $\tau \geq \frac{1}{\text{poly}(n, \frac{1}{\epsilon})}$ (to ensure tolerance isn't very small)



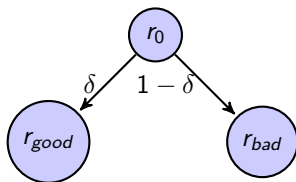
Proof Outline

for some $\tau > 0$, and $t \geq \tau$ CSQ oracle returns:

1 if $E_D[f(x)h(x)] \geq t + \tau$

0 if $E_D[f(x)h(x)] \leq t - \tau$

0 or 1 otherwise



- mutation pool size $p = O(\frac{\log 1/\delta}{\delta})$
- sample size $S = O(\frac{\log 1/\delta}{\tau^2})$

Thanks!